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Research Article

Effect of virtual reality meditation on sleep quality of intensive care unit patients: A randomised controlled trial

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ABSTRACT

Objective: This study aimed to investigate the effect of virtual reality meditation on sleep quality of intensive care unit patients.
Methods: This randomised controlled trial included 48 cardiac intensive care unit patients in a university hospital in Korea randomly allocated to the experimental (24) and the control group (24). For the experimental group, meditation was provided for 30 minutes using a head-mounted display for virtual reality, on the evening of the admission day.
Main outcome measures: The sleep quality of both groups was measured by self-report using Sleep Scale A and the activity tracker FitBit Charge 2.
Results: The experimental group reported significantly higher subjective sleep quality than did the control group. Activity tracker assessment indicated that total sleep time and light sleep time did not differ between the groups. However, the awake time was shorter, deep sleep time was longer and sleep efficiency was significantly higher in the experimental group than in the control group.
Conclusion: Virtual reality meditation positively affected the sleep quality of intensive care unit patients. Critical care nurses should consider using virtual reality meditation as a nursing intervention to improve the patient's sleep quality.

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Implications for clinical practice

- Virtual reality devices can be used in intensive care unit environments.
- A 30-minute virtual reality meditation may improve the subjective and objective quality of sleep of intensive care unit patients.
- Virtual reality meditation can be used as a sleep intervention for intensive care unit patients.

Introduction

Sleep quality tends to be poor in intensive care unit (ICU) patients. In addition to the anxiety and stress caused by illness and hospitalisation, noise and bright lighting in the ICU, and ongoing treatment and nursing activities make it difficult for the patient to sleep (Chahraoui et al., 2015; Mattiussi et al., 2019). Admission to the ICU results in an overall change in sleep architecture, including a higher proportion of light sleep, a relatively lower rate of deep sleep and sleep fragmentation (Elliott et al., 2013; Pulak and Jensen, 2016). These sleep disorders can negatively affect cardiovascular, respiratory, metabolic and immune functions, delay recovery, prolong hospital stay and even cause persistent sleep problems and poor quality of life after discharge (Altman et al., 2017; Beltrami et al., 2015).

To improve sleep quality of ICU patients, various interventions including noise and lighting reduction, eye mask, earplugs, massage, meditation, relaxation therapy, footbath, music, acupuncture and aromatherapy have been tried (Hu et al., 2015; Younis et al., 2019). Among these, meditation induces sleep by relaxing the

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mind and body (Black et al., 2015). Meditation is the process of relaxing and restoring the mind and induces complex and dynamic changes in brain activity (Escrichs et al., 2019). In this state of relaxation, the parasympathetic nerves are activated to lower blood pressure, pulse rate and to slow respiration, so to increase the rate of deep sleep during NREM sleep (Goldstein et al., 2019). A recent meta-analysis (Rusch et al., 2019) reported that meditation interventions significantly improved sleep quality at post-intervention and follow-up. According to the metacognitive model (Ong et al., 2012), meditation shifts mental and physical arousal into mental processes, which eventually leads to sleep through flexible thinking and peace of mind. Lau et al. (2018) also suggested that awareness and acceptance could be the mechanism by which meditation improves sleep quality.

The effect of meditation on sleep has been reported mainly in community-dwelling adults (Black et al., 2015; Goldstein et al., 2019). One reason for this is that meditation requires an environment in which to concentrate and a quiet space where you can focus on your mind with minimal body movement (Thomas and Cohen, 2014). However, the constant noise and lighting make it difficult to find a meditative space in the ICU, so research on meditation for ICU patients is limited.

Virtual Reality (VR) can be considered an alternative to a meditation space in the ICU. The VR experience, free from time and space, can allow patients to forget that they are in the ICU (Vazquez et al., 2013) and provide various content such as games, music, rehabilitation exercises, cognitive therapy and meditation (Kazzi et al., 2018). Gerber et al. (2017) reported that vital markers of physical stress, including heart rate and blood pressure, were reduced after providing 37 ICU patients with a nature-friendly VR visuo-acoustic stimulation.

Although research on VR as a treatment intervention is on the rise, studies in ICU patients are limited. To the best of our knowledge, no studies have attempted VR meditation in ICU patients. We hypothesised that VR meditation would allow ICU patients to relax their mind and body by blocking external stimuli and promote internal focus. The purpose of this study was to investigate the effects of VR meditation on the sleep of ICU patients to assess its potential as a sleep intervention. We hypothesised: 1) The subjective sleep quality of the experimental group will be higher than the control group and 2) The objective sleep quality including total sleep time, wake after sleep onset, light sleep time, deep sleep time and sleep efficacy of the experimental group will be higher than the control group.

Methods

Design and setting

This randomised controlled trial was conducted in the cardiac ICU of Dong-A university hospital from December 2018 to March 2019. The ICU had a total of 19 beds and mainly admitted cardiovascular patients. The average length of stay in 2018 was 2.3 days.

Participant selection

The participants were adults older than 18 years admitted to the ICU with cardiovascular disease, such as angina pectoris, myocardial infarction, heart failure, and cardiomyopathy. The exclusion criteria were: sedation with a RASS score <-1; visual or hearing difficulty for VR use; persistent pain or newly onset chest pain; use of muscle relaxants, sedatives, analgesics or hypnotics after admission; organ support treatments including mechanical ventilation, extracorporeal membrane oxygenation, continuous renal replacement therapy and intra-aortic balloon pump; and those on contact precautions for infection control.

The sample size was estimated using the G power 3.1.9.2 programme. When the effect size (d) was set to 0.89 based on two previous studies on meditation (Lee, 2013; Park and Choi, 2016), the significance level to (α) = 0.05 and the power to (1-β) = 0.8, the minimal total sample size was 42. We recruited 52 cardiac ICU patients considering a 20% dropout rate.

We posted a public announcement for the recruitment of study participants on a bulletin board at the entrance of the ICU. The announcement included the title of the study, participants, procedure, and rewards. When a patient or family inquired about participation, the ICU nurses gave instructions on how to participate in the study. One of the researchers (SYL) confirmed that the applicants met the inclusion criteria. Applicants who met the criteria and had provided written informed consent were randomly allocated to experimental or control groups according to the 1:1 random list order created by a research assistant with the Random Allocation Software programme version 2.0.0. The random list was kept in a password-set file by the assistant, so that the researchers and ICU nurses did not know which group the next participant would be assigned to.

Ethical approval

This study was approved by the Institutional Review Board (IRB) of the Dong-A University Hospital (approval number: DAUHIRB-18-239).

Data collection

Intervention: VR meditation

VR meditation was provided using the NOON PRO (FXGear, Inc., Seoul, Korea: Price $200, £150). This is a head-mounted display (HMD) designed to enclose both eyes and ears and block external stimuli almost entirely. We uploaded the "Relaxing Music for Meditation" programme, which was designed to induce meditation and consists of eight meditation-videos of 30 minutes.

When a participant selects a meditation video wearing the HMD, natural scenes such as mountains, sea or lakes appear on the VR screen, and background sounds and calm music allow participants to immerse themselves in the experience. For example, when a participant selects a beach, the sea and calm waves spread out in front of them and peaceful sounds of waves and gulls flow from built-in speakers. The 360 degree landscape turns with the gaze of the participant, creating the feeling of being inside the landscape.

Procedure

The study began on the day of admission to the ICU. Once the group was allocated, all participants completed the Pittsburgh Sleep Quality Index (PSQI) questionnaire on their usual sleep and the activity tracker band was placed on their wrist to record their movements. Additionally, the HMD was provided to the experimental group to choose the videos in advance and practise wearing and using the equipment. It took about 20 minutes for the experimental group to learn how to use the HMD. Double blindness was not possible due to the interventional characteristics of the study using VR equipment.

The experimental group used the VR meditation programme for 30 minutes before bedtime (9–11 PM) on the day of ICU admission. Participants could sleep while wearing the device if desired. The control group was provided with daily routine sleep intervention

according to ICU protocol (eye mask or earplugs if desired). All participants wore the activity tracker during sleep.

The following morning, 30 minutes after waking, both groups completed a questionnaire to assess subjective sleep quality. The objective sleep quality data such as total sleep time, wake after sleep onset, light sleep time and deep sleep time was downloaded from the activity tracker.

Measurements

Usual sleep quality

To verify the pre-interventional homogeneity of the experimental and control groups, usual sleep quality was assessed using the Pittsburgh Sleep Quality Index (PSQI) developed by Buysse et al. (1989) and translated and validated by Sohn et al. (2012) to Korean. The PSQI is a self-reported questionnaire that measures sleep quality over the past month and consists of 19 items. The total score for the PSQI is 0 – 21; the lower the score, the higher the quality of sleep. Reliability reported by Sohn et al. (2012) was Cronbach's α = 0.84, and Cronbach's α = 0.70 in the present study.

Subjective sleep assessment

Subjective sleep quality after the intervention was measured using the 'Korean Sleep Scale A' developed and validated by Oh et al. (1998). The scale consists of 15 items and four factors: sleep patterns, evaluation, results, and cause of the disturbance. This scale is a 4-point Likert type (Strongly disagree = 1, Disagree = 2, Agree = 3, Strongly agree = 4), the higher the score, the higher the quality of sleep. The reliability reported by Oh et al. (1998) was Cronbach's α = 0.75 and the reliability in the present study was Cronbach's α = 0.77.

Objective sleep assessment

Objective quality of sleep was measured using the activity tracker FitBit Charge 2 (FB407, FitBit® Inc., San Francisco, California, USA), a 3-D micro-electro-mechanical system that uses body movement and heart rate variability sensors. FitBit is a wearable wristband that synchronises with a mobile application or computer dashboard. The device estimates sleep for each 30 s period by integrating a participant's movement and heart rate data. The - FitBit provides sleep stage data including sleep time, wake, light sleep, deep sleep, and rapid eye movement (REM) sleep (Fig. 1). We retrieved total sleep time (TST), wake after sleep onset (WASO), light sleep time, and deep sleep time from the android application of FitBit. In the case of WASO, the frequency and time of awakening were provided. We analysed the time of WASO. Sleep efficiency (SE) was calculated as the ratio of deep and light sleep time to TST. According to a recent meta-analysis (Haghayegh et al., 2019), sleep stage models such as FitBit Charge 2 showed no significant differences in WASO, TST, and SE measurements compared to polysomnography (PSG), and acceptable sensitivity and specificity values in detecting sleep epochs.

Data analysis

The collected data were analysed using SPSS Win 22.0 programme (IBM Inc., Armonk, NY). Participants' characteristics are presented using descriptive statistics included mean (M) and standard deviation (±) for continuous variables and frequency and percentage for categorical variables. The normality of each variable was verified using the Kolmogorov-Smirnov test. The homogeneity test of experimental and control groups and the hypothesis test were analysed using the chi- square test ($\chi 2$), *t*-test (t) and Mann Whitney *U* test (U) for non-normal variables. All significance levels were set to α <0.05.

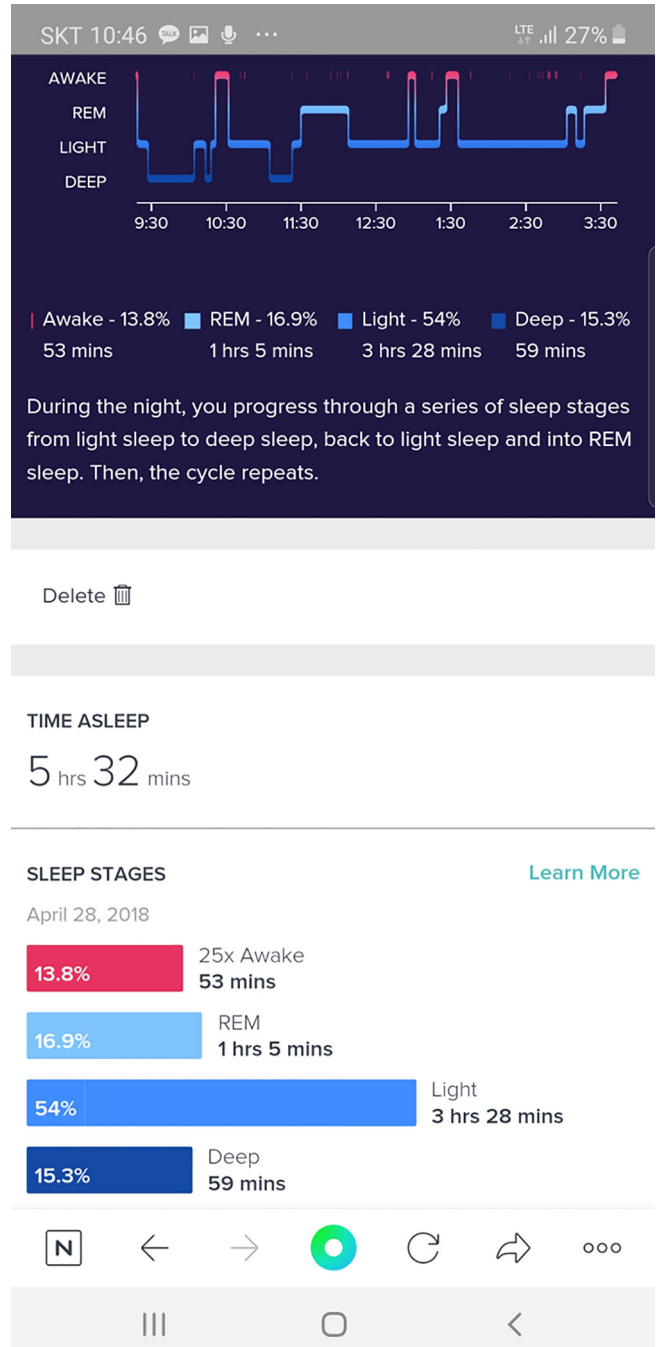

Fig. 1. Screenshot of sleep data provided by FitBit Charge 2.

Results

Participant characteristics and homogeneity test

Out of 64 volunteers willing to participate in the study, 12 were excluded because they did not meet the selection criteria. We randomly assigned the remaining 52 to the experimental and control groups (1:1). Two participants in the experimental group withdrew from the study due to the inconvenience of wearing the

HMD device. One of them said that the HMD itself was heavy and the other was annoyed because he was not use to using the device. One participant in the control group rejected the post-measurements and another dropped out due to chest pain. As a result, a total of 48 ICU patients participated in the study (Fig. 2).

The mean age of study participants was 66.42 years. Most of them were male and married and about half were diagnosed with myocardial infarction. Their mean APACHE II score was 14.31, ICU days was 2.79, and hospital days was 10.83. There were no significant differences in general and clinical characteristics of experimental and control groups. In terms of usual sleep quality, there was no significant difference in PSQI scores between the groups (t = -0.69, p = .494) (Table 1).

Sleep quality after the intervention

The mean subjective sleep quality score of the experimental group with VR meditation after the intervention (2.25 ± 0.19) was significantly higher than that of the control group (2.06 ± 0.21) (t = 3.34, p = .002). The scores of four factors of Sleep Scale A (sleep pattern, sleep evaluation, sleep result and cause of sleep disturbance) were higher in the experimental group (Table 2).

The mean total sleep time of the experimental group after the intervention was 390.83 ± 31.09 minutes and that of the control group was 387.92 ± 30.65 min, with no significant difference between the groups (t = 0.33, p = .745). The median (Mdn) and interquartile range (IQR) wake after sleep onset of the

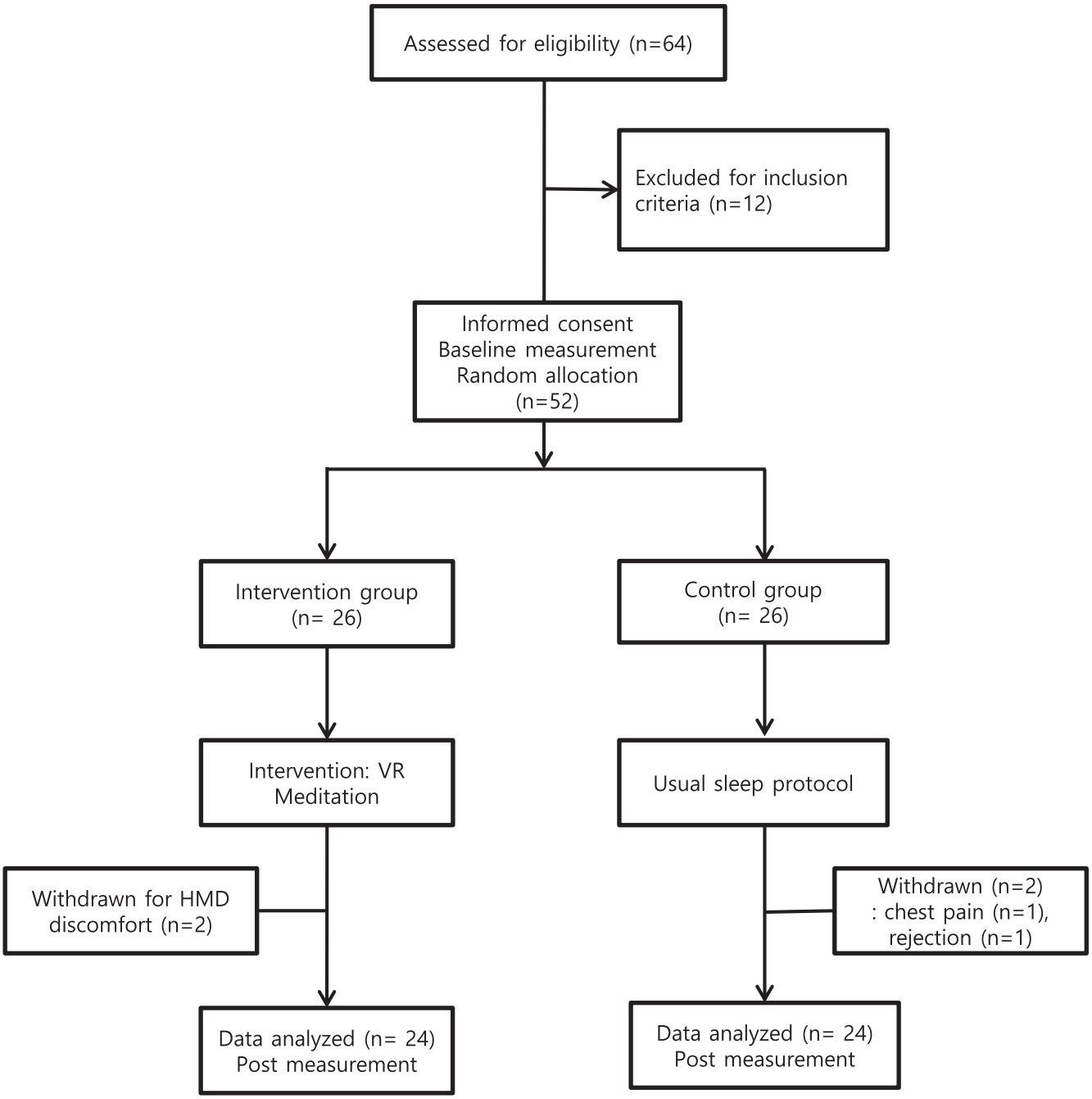

Fig. 2. Flow diagram of the study.

Table 1
Homogeneity test between experimental and control group (N = 48).

Variable	Category	Total (n = 48) N(%)/M ± SD	Experimental (n = 24) n(%)/M ± SD	Control (n = 24) n(%)/M ± SD	Statistic test χ^2/t	Significance *p*
Gender	Male	32 (66.7)	15 (62.5)	17 (70.8)	0.38	0.540
	Female	16 (33.3)	9 (37.5)	7 (29.2)		
Age (years)		66.42 ± 16.51	69.46 ± 16.27	63.38 ± 16.53	1.29	0.205
Marital status	Single	13 (27.0)	5 (20.8)	8 (33.3)	0.95	0.330
	Married	35 (73.0)	19 (79.2)	16 (66.7)		
Religion	No	32 (66.7)	15 (62.5)	17 (70.8)	0.38	0.540
	Yes	16 (33.3)	9 (37.5)	7 (29.2)		
Diagnosis	Congestive heart failure	6 (12.5)	5 (20.8)	1 (4.2)	11.57	0.315
	Cardiomyopathy	2 (4.2)	0 (0)	2 (8.3)		
	Myocardial infarction	26 (54.2)	13 (54.1)	13 (54.1)		
	Others	12 (25.1)	6 (25.0)	8 (33.4)		
APACHE II score		14.31 ± 5.03	14.50 ± 4.7	14.13 ± 5.43	0.26	0.799
ICU days		2.79 ± 2.14	2.71 ± 2.37	2.87 ± 1.94	−0.27	0.791
Hospital days		10.83 ± 8.48	10.63 ± 9.43	11.04 ± 7.62	−0.17	0.867
PSQI score		8.96 ± 2.50	8.71 ± 2.33	9.21 ± 2.69	−0.69	0.494

PSQI = Pittsburgh Sleep Quality Index.

Table 2
Comparison of subjective sleep quality after intervention between experimental and control group (N = 48).

Sleep Scale A	Experimental (n = 24) Mdn (IQR) M ± SD	Control (n = 24) Mdn (IQR) M ± SD	Statistic test t/U	Significance *p*
Sleep pattern	2.20 (0.40)	2.00 (0.65)	−1.95*	0.051
Sleep evaluation	2.13 (0.50)	2.00 (0.25)	−2.97*	0.003
Sleep result	2.00 (1.00)	2.00 (0)	−2.12*	0.034
Cause of sleep disturbance	2.50 (0.62)	2.25 (0.50)	−3.03*	0.002
Total	2.25 ± 0.19	2.06 ± 0.21	3.34	0.002

*Mann-Whitney *U* test.

experimental group was 52.50 (12.25) minutes and that of the control group was 65.00 (15.50) minutes. There was a significant difference between the groups (U = -2.61, p = .009). The median (Mdn) and interquartile range (IQR) light sleep time of the experimental group was 109.00 (33.35) minutes and that of the control group was 128.00 (38.50) minutes, with no significant difference between the groups (U = −1.81, p = .071). The mean deep sleep time of the experimental group was 75.83 ± 17.03 minutes and that of the control group was 63.92 ± 12.09 minutes, which was significantly different between the groups (t = 2.80, p = .008). The mean sleep efficacy of the experimental group was 85.68 ± 3.38, which was significantly higher than the control group, 83.36 ± 3.20 (t = 2.44, p = .018) (Table 3).

Discussion

We investigated the effects of VR meditation on the sleep of ICU patients. The 30-minute VR meditation given to cardiac ICU patients in the evening of the admission day improved subjective sleep quality, WASO, deep sleep time, and SE. However, VR meditation did not improve TST and light sleep time.

The TST of the present participants, measured by the activity tracker was 389 minutes, similar to the 396 minutes reported by Kroll et al. (2017) using the same device. However, this is about 90 minutes longer than TST reported in another study (Elliott et al., 2013) that assessed the sleep patterns of ICU patients using 24-hour PSG. This difference in TST can be explained by the different methods used. Unlike PSG, activity trackers are motion-based and thus tend to overestimate the TST of ICU patients who are primarily in bed (Beecroft et al., 2008; De Zambotti et al., 2016).

Polysomnography, which detects electroencephalography, electromyography, electrooculography, electrocardiography and oxygen saturation changes remains the most accurate sleep assessment method (Medrzycka-Dabrowska et al., 2018). However, PSG has limited application in clinical studies because it is expensive and requires expert interpretation and specific equipment and space (Younis et al., 2019). Activity trackers, however, measure

Table 3
Comparison of objective sleep quality after intervention between experimental and control group (N = 48).

Activity tracker records	Experimental (n = 24) Mdn (IQR) M ± SD	Control (n = 24) Mdn (IQR) M ± SD	Statistic test t/U	Significance *p*
TST (min)	390.83 ± 31.09	387.92 ± 30.65	0.33	0.745
WASO (min)	52.50 (12.25)	65.00 (15.50)	−2.61*	0.009
Light sleep time (min)	109.00 (33.35)	128.00 (38.50)	−1.81*	0.071
Deep sleep time (min)	75.83 ± 17.03	63.92 ± 12.09	2.80	0.008
SE (%)	85.68 ± 3.38	83.36 ± 3.20	2.44	0.018

*Mann-Whitney *U* test, TST = total sleep time, WASO = wake after sleep onset, SE = sleep efficiency.

sleep and wake-up times by detecting body movements with a small device worn on the wrist. Activity trackers are non-invasive, easy to use, relatively inexpensive and are commonly used in sleep research (Beltrami et al., 2015; Evenson et al., 2015; Lee et al., 2014). The device we used has high sensitivity and moderate specificity as a sleep assessment, as well as feasibility and wearability in a study of ICU patients (Kroll et al., 2017). However, the activity tracker's specificity for measuring the sleep quality of patients staying in the ICU for a short time remains questionable (Beecroft et al., 2008). We assessed subjective sleep quality of participants to supplement these limitations.

Regarding sleep depth, the light sleep time of our participants was nearly twice as long as the deep sleep time. This finding is consistent with a previous study (Elliott et al., 2013) in which light sleep increased in ICU patients. Systematic reviews of sleep in ICU patients (Beltrami et al., 2015; Medrzycka-Dabrowska et al., 2018) reported that light sleep times were longer than deep sleep, and WASO was long. The present participants' WASO was also long, 55–64 minutes. This is consistent with the results of a previous study (Beltrami et al., 2015) in which ICU patients' sleep was fragmented. During the present study, participants' vital signs were measured every hour, which might have increased WASO. Once again, the ICU nurses need to adjust the night-time activity schedule so that the patient's sleep is not interrupted.

In our study, VR meditation for 30 minutes improved subjective sleep quality, WASO, deep sleep time and SE in ICU patients. According to the 2017 National Health Interview Survey, meditation was the first choice for Americans with sleep disorders, especially young and educated individuals and those with heart disease tended to use mind-body therapies, including spiritual and mindfulness meditation (Voib et al., 2019). Meditation is not only effective for a trained person. In a study (Dentico et al., 2016) that compared novice and trained meditators, EEG activity corresponding to deep sleep stages was activated by short meditation training, and there was no difference in the degree of EEG activation between the two groups. Given that patients who are unfamiliar with meditation can improve sleep through this technique, it should be considered a valid nursing intervention.

The results of this study suggest the possibility of using VR space in the ICU. Wearing a VR HMD can block ICU patients from surrounding noise and scenes that may cause anxiety. The use of virtual space may be able to provide a variety of interventions for ICU patients more efficiently. Vazquez et al. (2013) reported that exploring a variety of environments using VR reduced postoperative anxiety in cardiac recovery ICU patients. VR presentation of cognitive stimuli, such as peaceful landscapes, has been reported to stabilise the vital signs and eye movements of ICU patients (Gerber et al., 2017). VR-based neurocognitive interventions are also applicable to ICU patients undergoing mechanical ventilation and have the effect of relaxing them without fatigue (Turon et al., 2017). Sensory deprivation or noxious environmental stimuli in the ICU are associated with poor cognitive function and quality of life in survivors after discharge (Hofhuis et al., 2008; Timmers et al., 2011). Also, sleep quality in ICU is a major cause of sleep disorders after discharge and sleep is very important for the physical and psychological well-being of individuals (Altman et al., 2017). Overall, VR meditation has the potential to contribute to improving the long-term outcomes of ICU survivors.

Study limitations

First, this study was performed in a cardiac ICU of a university hospital, which limits the generalisability of the findings. Because of the nature of the intervention itself, we excluded sedated or ventilated patients. Although our participants' average APACHE scores, ICU days and hospital days were similar to those of Medical Information Mart for Intensive Care (MIMIC)-III big data (Johnson et al., 2016), it is not known whether the results of this study can be applied to ICU patients with higher severity. Second, sleep was measured for only one night, so we could not investigate the difference in sleep quality and the effects of intervention during the ICU stay. Therefore, follow-up studies are needed to evaluate continuous sleep over the ICU admission period. Third, we did not consider environmental changes such as daily noises, odour, and lighting in the ICU. Finally, it was not possible to determine whether our VR meditation brought the participants to an actual meditation state; further research is needed, as the exact mechanism of meditation remains unclear (Wahbeh et al., 2018). Therefore, it is necessary to reinvestigate the effects of VR meditation by designing rigorous experimental studies while controlling factors affecting ICU sleep.

Conclusion

VR meditation improved subjective sleep assessment, reduced awakening time, and increased deep sleep time and sleep efficiency of ICU patients. Thus, VR is useful as a nursing intervention method to provide a positive cognitive stimulus and block environmental stimuli in the ICU. We recommend using VR meditation as a critical care nursing intervention given its potential to improve sleep quality in ICU patients. Additionally, the feasibility and generalisability of VR interventions need to be established through more rigorous clinical trials that control several influencing factors affecting sleep in the ICU.

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Authors' Contributions

JK contributed to the study conception and design. SYL was responsible for data collection and statistical analyses. Both authors drafted the manuscript, and carried out critical revisions of the paper.

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